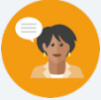


10.7 Solving Problems Involving Intersecting Chords

Lesson Introduction

Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.		Learning Target	I can solve problems involving chords of a circle and their related angles.	
Benchmark Coherence	Building On		Working Toward		
	N/A		N/A		
Essential Questions	- How can I use what I know about chords of a circle to solve problems? - What is the relationship between intersecting chords and the angles in the intersecting chords?				
Lesson Narrative	In this lesson, students learn about angles in intersecting chords and how the relationships that exist in each can be used to solve problems. Students begin by exploring intersecting chords to discover the angle relationships that exist in intersecting chords. They will use this information to solve various types of problems throughout the lesson.				
Component Summary	10.7.1 Warm-Up (~5 min.)	Evaluate the size of sectors drawn in a circle (<i>SE p. 154</i>)	10.7.4 Guided Practice (5-10 min.)	Solve problems involving angles in intersecting chords (<i>SE p. 157</i>)	
	10.7.2 Exploration (5-10 min.)	Examine chords of a circle (<i>SE p. 154</i>)	10.7.5 Your Turn! (5 min.)	Solve a problem involving angles in intersecting chords (<i>SE p. 158</i>)	
	10.7.3 Collaboration (10-15 min.)	Discover the relationship between the measures of angles of intersecting chords and their intersected arcs (<i>SE p. 155</i>)	10.7.6 Wrap-Up (~5 min.)	Reflect on the level of understanding of the lesson (<i>SE p. 159</i>)	
Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Chord - Angles of intersecting chords theorem <u>Additional Glossary:</u> - Vertical angles		- Protractor (Optional) Computers for GeoGebra (Optional) Graphing utility		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recall the relationship of vertical angles. - Students may inaccurately label a perpendicular bisector when it doesn't meet the conditions of one.	Consider adding a simple problem with vertical angles as a quick refresher, so that it is at the front of their mind prior to 10.7.3 Collaboration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder During the 10.7.1 Warm-Up, students are presented with a circle graph about student hours by subject in school. Students will be asked to share what they notice about the circle graph to provide a low entry point for all students.	MA.K12.MTR1.1: Participate in Effortful Learning The primary set of activities in this lesson are Collaboration and Exploration activities. Throughout these activities, students will build perseverance by modifying their methods, as needed, while solving a challenging task. They will work to help and support each other when attempting each problem.
 Differentiate Instruction	For 10.7.2 Exploration, students may need additional scaffolding to see the relationship between the angles and the intercepted arcs. Consider the use of sentence stems to build toward this discovery. Consider allowing students the opportunity to use GeoGebra as a resource for investigating angle measures of intersecting chords. Additional differentiation is notated throughout the lesson guide.	
Lesson Notes:		

10.7.1 Warm-Up: Sectors

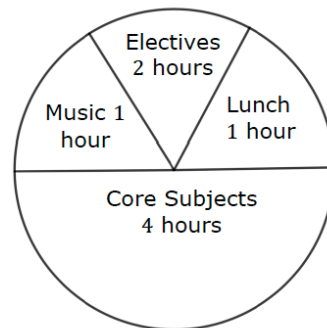
Component Overview: Students will answer three questions about sectors from a provided circle graph, in context.



10.7.1 Warm-Up: Sectors

1. What do you notice about the circle graph?

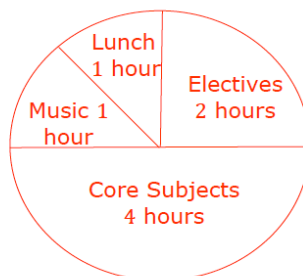
Answers will vary. I noticed that the number of hours in each sector aren't accurately represented by the sizes of each sector.



2. Identify a sector that is incorrectly sized.

Answers will vary. Music represents one hour, however, Electives represents two hours, but they are both the same sector size.

3. Redraw the circle graph so that the sizes are proportional to the number of hours.



For this activity, students are provided with a low-entry question first. This allows all students to actively engage in the content, regardless of their background. Provide some individual think time, and then allow students to share their thoughts with a partner before a whole-group review of the answers.

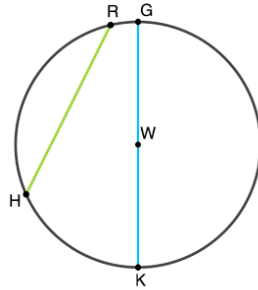
10.7.2 Exploration: Chords

Component Overview: Students will explore the concept of chords using a single example with three scaffolded questions.



10.7.2 Exploration: Chords

Circle W is shown where points R , G , K , and H lie on the circle.



1. What similarities do you notice between line segment RH and line segment GK ?

Answers will vary. Both line segments have points on the circle, both line segments are in the circle.

2. What differences do you notice between line segment RH and line segment GK ?

Answers will vary. Line segment GK goes through the center, line segment GK is longer.

A **chord** is a line segment that connects two points on the curve of a circle.

3. Complete the statements.

Line segment RH is
of the circle.

- ☐ a diameter
- ☒ a chord
- ☐ both a diameter and a chord

Line segment GK is
of the circle.

- ☐ a diameter
- ☐ a chord
- ☒ both a diameter and a chord



Consider making a list of students' answers to questions 1 and 2 for all to see. Students benefit from hearing the thinking of their peers.



- *What is the relationship between a diameter and a chord?*
- *How can a diameter also be classified as a chord?*

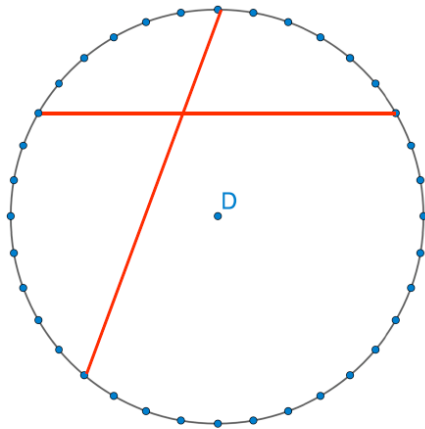
10.7.3 Collaboration: Angles in Intersecting Chords

Component Overview: Students will learn about the angles that exist when chords intersect. For this activity, students should work with a partner.



10.7.3 Collaboration: Angles in Intersecting Chords

1. With your partner, complete the following using Circle D.
 - a. Draw and label two chords that intersect inside the circle, making sure the endpoints fall on the points. When drawing the chords, do not draw the same chords as your partner and avoid drawing diameters.



- b. Choose one of the angles to measure and label. Then, complete the table.

	Angle Measure	Measure of Arc it Intercepts	Measure of the Arc the Vertical Angle Intercepts
Your Angle	120°	70°	170°
Your Partner's Angle	60°	30°	90°



For this activity, students will need a partner. Try and pair students together based on their performance throughout Unit 10. This can allow more students to get assistance during the facilitation of this activity, as well as learn with their peers.



If students struggle finding the measure of the arcs, consider asking the following:

- What is the degree measure of a whole circle?
- What do you notice about the dots on the circle? How many are there?
- What is the degree measure between each dot on the circle?



For question 1 part b, students will need to use their pre-requisite knowledge of vertical angles. Watch for students who may get this incorrect to determine if a refresher on vertical angles may be needed. Use this opportunity to spiral back to the relationship between the measures of vertical angles. Make connections that an angle and its vertical angle will have the same relationship with the intercepted arcs.

10.7.3 Collaboration: Angles in Intersecting Chords (continued)

c. Discuss the following with your partner.

- What do you notice about the values in the table?
- What do you think the relationship is between the angle measure and the two intercepted arcs?

Answers will vary. Our angle measure falls exactly halfway between the values of the two arc measures. An angle made by two intersecting chords is the average of the measures of the two arcs intercepted by the chords.

d. Write a formula that would generalize your discovery from the discussion you had in Part C.

$$m\angle A = \frac{m(\text{arc } 1) + m(\text{arc } 2)}{2}$$

e. Complete the statement.



Angles of Intersecting Chords Theorem

The angle formed inside of a circle by two intersecting chords is
☐ double
☐ equal
☒ half
 the sum of the measures of the arcs intercepted by the angle and its vertical angle.



Listen for student conversations when circulating the room. Ensure that the conversation for question 1 part c stays centered on the questions that go along with it, and not general noticing. The intent is for what they notice to serve as the springboard for their formula in question 1 part d. If needed, consider offering scaffolding support in the form of sentence stems:

- The sum of the measure of the two arcs an angle and its _____ intercepts is _____.
- The relationship between the _____ of the two arcs and the angle measure is _____.



For students that finish early, have them make connections between the discovery here to the measure of a central angle and its intercepted arc.

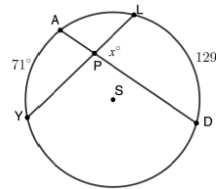
10.7.4 Guided Practice: Solving Problems Involving Angles in Intersecting Chords

Component Overview: Students will complete three practice problems on solving real-world and mathematical problems involving intersecting chords.

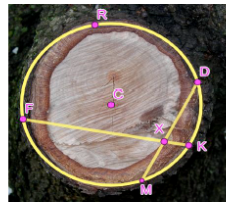


10.7.4 Guided Practice: Solving Problems Involving Angles in Intersecting Chords

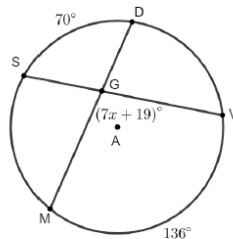
- Circle S is shown, where chords $\overline{AD}$ and $\overline{YL}$ intersect at point P . Find the value of x for $\angle LPD$, given $m\widehat{AY} = 71^\circ$ and $m\widehat{LD} = 129^\circ$.
 $x = 100$



- A tree is cut down and a circular tree stump is left. The top of the tree stump has two cracks that represent two chords, $\overline{FK}$ and $\overline{DM}$, intersecting at point X in Circle C . If $m\widehat{FRD} = 175^\circ$ and $m\widehat{KM} = 39^\circ$, what is the measure of $\angle FXD$?
 $m\angle FXD = 107^\circ$



- Circle A is shown, where chords $\overline{SV}$ and $\overline{MD}$ intersect at point G . Find the value of x , given $m\widehat{SD} = 70^\circ$, $m\widehat{MV} = 136^\circ$, and $m\angle MGV = (7x + 19)^\circ$.
 $x = 12$



For this activity, consider using these three questions as a gradual release practice:

For question 1, elicit information from students about what they notice in the circle. Use that information to connect students to the correct markings, and then model the correct thinking for solving.

For question 2, release students to try it on their own and then check with a partner before reviewing together as a class. Struggling readers may need additional processing time.

For question 3, fully release students to try this one on their own from start to finish before bringing the class back together to review the answer.



If students finish early, challenge them to name one piece of information that could be included that would allow them to find the measure of $\widehat{SM}$ and $\widehat{DV}$.

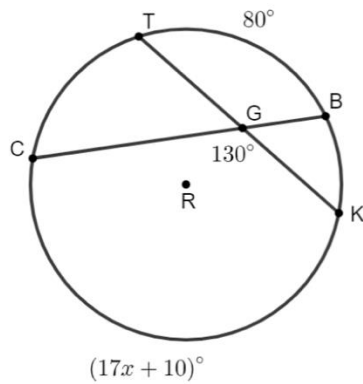
10.7.5 Your Turn!

Component Overview: Students will solve one mathematical problem using angles of intersecting chord relationships.



10.7.5 Your Turn!

1. Circle R is shown, where chords $\overline{TK}$ and $\overline{CB}$ intersect at point G . Find the value of x , given $m\widehat{TB} = 80^\circ$, $m\widehat{CK} = (17x + 10)^\circ$, and $m\angle CGK = 130^\circ$.



$$x = 10$$



For this question, students should complete it independently. This will provide valuable data to determine whether students are grasping the content from this lesson, and where they may be getting stuck.



For students who finish early, consider having them determine $m\widehat{CK}$.



10.7.6 Wrap-Up: Reflection

Answers will vary.

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this, |
| I can't get started. | I need more practice. | and I feel confident. |

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this, |
| I can't get started. | I need more practice. | and I feel confident. |

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this, |
| I can't get started. | I need more practice. | and I feel confident. |



The information provided from the student here, coupled with what was observed during the lesson, can provide information for differentiating instruction in later lessons.